

# Roy (Yuanxi) Cheng

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## EDUCATION

### University of California, Los Angeles (UCLA)

Sep 2023 – Mar 2027

*Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering*

*GPA: 3.75*

- **Awards & Activities:** Dean's Honor's List, Data Science Union, 180 Degrees Consulting, Pi Sigma Epsilon
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Data Analysis & Regression, Machine Learning, Neural Networks & Deep Learning, Data Mining, Data Structures & Algorithms, Probability & Statistical Inference, Databases, NLP, Monte Carlo, Optimization

## WORK AND LEADERSHIP EXPERIENCE

### Capital One

McLean, Virginia

*Product Management and Analysis Intern*

*Jun 2026 – Present*

- Analyze corporate workforce data with SQL and Python, identifying effects and factors tied to successful org transitions
- Synthesize insights into Tableau dashboard of key reorg metrics, presenting to SVP to help HRBPs make reorg decisions
- Process 1.5k BA performance reviews through a semantic similarity pipeline, reducing 923 O\*NET occupations to 200+ DWA candidates via FAISS and ranking to 31 skills by prevalence and LLM evidence score
- Build composite scoring model ranking skills by frequency and evidence strength, delivering a recruiter-facing insights dashboard from bottom-up analysis of 1.5k performance reviews

### Kohl's

Los Angeles, California

*Project Manager and Data Analyst*

*Jun 2025 – Mar 2026*

- Analyzed 8.5M+ logistics records with SQL and Python, resolving data quality to build a reliable shipment-level dataset
- Conducted EDA in Vertex AI Workbench, engineering supply-side shipment features that sharpened transit time forecasts
- Developed two-stage XGBoost forecasting model with time-aware validation, improving arrival prediction MAE by 83%

### ACTUM Digital

Prague, Czech Republic

*Data Analyst and AI Intern*

*Jun 2025 – Aug 2025*

- Spearheaded project tasks in Jira, presenting structured updates to manager that accelerated delivery timelines by 20%
- Engineered agentic workflow with Python to denoise and extract Confluence data, generate metadata, and embed documents into a Milvusvector database for knowledgebase MCP server development
- Created AI assistant hub of context-trained LLMs with an MCP server integration, co-presenting live demo to Directors

### Million Dollar Baby

Los Angeles, California

*Data Analysis Consultant*

*Apr 2025 – Jun 2025*

- Analyzed \$10M+ in advertising spend, surfacing efficiency insights that inform growth and marketing investment strategy
- Presented CAC and RoAS assessments across multiple day windows with Python, guiding decisions on spend allocation
- Built Excel dashboard tracking 20+ marketing KPIs, giving managers a recurring view of acquisition and return trends

## PROJECTS AND RESEARCH

### COVID Pandemic Effects on Secondary Education | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned 100k+ student records across multi-level datasets with pandas, enabling accurate longitudinal analysis in Tableau
- Aggregated 4 years of test scores into time-series visualizations, revealing student performance impacts tied to pandemic
- Created interactive Tableau map linking geographic data to score changes across U.S. states for a public research website

### Adrenaline and Cognitive Fatigue Effects on Athletic Performance | *R, ggplot2, tidyverse, dplyr*

- Designed 2<sup>3</sup> factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at  $\alpha = 0.05$
- Validated ANOVA assumptions using log transforms and ggplot2 diagnostics, confirming normality and homoscedasticity

### Biomedical Tissue Imaging | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article evaluating CNN-based instance segmentation architectures with AP metrics, benchmarking cytoplasm boundary precision for biomedical imaging
- Converted 4k+ CytoNuke annotations to COCO format, standardizing training across 3 segmentation architectures
- Trained YOLOv8 instance segmentation models in PyTorch, improving cytoplasm AP50 by 5% over Cyto-RCNN baseline

## TECHNICAL SKILLS

**Languages:** SQL, Python, R, C++, JavaScript, Java

**Libs/Frameworks:** pandas, NumPy, Matplotlib, scikit-learn, XGBoost, MLflow, PyTorch, TensorFlow, LangChain, Keras

**Tools & Platforms:** Tableau, BigQuery, Snowflake, GCP, AWS, Vertex AI, Git, GitHub, Claude Code, Jupyter Notebooks